

Summary of Validation and Performance Experiments

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This section presents a comprehensive evaluation of the proposed skip-module architectures (A_1, A_2). We compare our frame-level skipping approach against standard lightweight alternatives (M_1, M_2) and the heavy baseline detector.

1 Component-Level Performance (Baselines)

Table 1 assesses the intrinsic recall and filtering capability of each method as a standalone component.

Table 1: Compare our BIG MODEL (no skip module) performance against other lightweight DL classifier or Object Detectors.

Method	Recall	False Alarm Rate	FPS
M1 (MobileNetV2 Classifier)	80.0%	10.0%	30
M2 (YOLOv8-Nano Detector)	85.0%	8.0%	45
BIG model	90.0%	5.0%	60

2 Hyperparameter Optimization (Validation Set)

Table 2 specifies the search space for the rule-based skip-module parameters. The ranking and selection of the optimal configuration (θ^*) based on the validation set results are detailed in Table 3.

Table 2: Hyperparameter search space and selected settings for both skip-module variants.

Group	Parameter	A1 (Baseline)	A2 (Proposed)	Note
Shared	Grid size ($N \times N$)	16×16	16×16	Balances spatial sensitivity and runtime cost.
Shared	Motion threshold	2.0	1.5	Lower threshold helps capture subtle smoke motion.
Fire color rule	Red threshold	N/A	selected	Detects bright flame-like red regions.
	Saturation threshold	N/A	selected	Helps reject non-fire bright objects.
	Intensity threshold	N/A	selected	Focuses on strong flame responses.
Smoke grayness rule	Grayness range	N/A	[0.7, 0.95]	Captures low-saturation semi-transparent smoke.
	Texture consistency	N/A	selected	Helps distinguish smoke from random motion.
Temporal rule	Window size (frames)	minimal	5	Suppresses transient noise and improves stability.

Table 3: Validation-set performance under the hard safety constraint Recall(Anomaly). Final model selected by maximizing SkipScore among feasible configurations.

Method	Configuration	Recall (Val)	Filter Rate (Val)	SkipScore	Role
A1 (Baseline)	16×16 2.0 motion only	97.2	65.0	N/A	Motion-only baseline – fails safety target (<99).
A2 (Proposed)	16×16 1.5 smoke rules	99.1	72.1	0.714	Selected final configuration.
A2 (Proposed)	16×16 2.0 smoke rules	99.0	68.3	0.676	Safe but less efficient.
A2 (Proposed)	32×32 1.0 smoke rules	99.2	65.7	0.652	Slightly higher recall lower efficiency.

3 End-to-End System Performance (Test Set)

The overall impact of integrating the skip modules into the full detection pipeline is quantified in Table 4 (frame-level accuracy/latency), also comparing against the baseline system without skipping.

Table 4: End-to-End System Performance Comparison on Test Set.

Config	Recall	False Alarm Rate	Accuracy	Skip Rate	FPS
BIG MODEL only	98.5%	50.0ms	312	0%	20
M1	76.0%	15.0ms	123123	70%	66
M2	81.5%	22.0ms	123123	56%	45
Ours (Appr. 1 + BIG)	98.3%	18.3ms	123	63%	54
Ours (Appr. 2 + BIG)	98.5%	16.5ms	3123123	67%	60

4 Comparison with Other Temporal Methods

Finally, Table 5 compares our method against other temporal processing techniques, evaluating both detection performance and computational efficiency.

Table 5: Comparison of Video-level performance metrics, measuring the delay and initial detection speed.

Method	Scope	Delay/Frame	Min Latency	First Alarm	Compute/Sec	Video Recall	False Alarm
M3 Temporal Voting	Video(30f)	25ms/f	30 Frames	>750ms	750ms	97.5%	1.2%
Ours (Appr.2+BIG)	Frame-level	16.5ms/f	1 Frame	52.5ms	495ms	98.4%	1.5%